Econ 5253 – Problem Set 12

Cheng

May 7, 2026

1–3. Repository setup and packages

The required commands (`git pull origin master`, `git fetch upstream`) were issued from the OSCER `DScourseS26` folder, and the R packages `sampleSelection`, `tidyverse`, and `modelsummary` were installed. All analysis is reproduced by the script `PS12_Cheng.R`.

4–6. Data summary and missingness

The data set `wages12.csv` contains $N = 2{,}229$ observations on working women in the U.S. in 1988. After converting `college`, `married`, and `union` to factors, Table 1 reports descriptive statistics.

Table 1: Summary statistics for the `wages12` data

Variable	Unique	Missing %	Mean	SD	Min	Max
`logwage`	1546	30.69	1.652	0.688	−0.956	4.166
`hgc`	14	0.00	12.455	2.444	5.000	18.000
`exper`	1932	0.00	6.435	4.867	0.000	25.000
`kids`	2	0.00	0.429	0.495	0.000	1.000
Categorical variables (counts and percentages)						
`college` = Yes	233 (10.5%)	(No: 1996, 89.5%)				
`married` = Yes	1415 (63.5%)	(No: 814, 36.5%)				
`union` = Yes	529 (23.7%)	(No: 1700, 76.3%)				

Discussion. The numbers look plausible for a 1988 cross-section of working women: average completed schooling is about 12.5 years (roughly a high-school graduate), average labour-market experience is 6.4 years, about two-thirds of the sample is married, and 23.7% are union members. Mean log wage is 1.65 (a wage of about \$5.20/hour, consistent with 1988 dollars).

Missingness. Log wages are missing for 684 of 2229 women, a **30.7%** missing rate. Because the sample only contains women who were working, the missingness in `logwage` is most likely **Missing Not At Random (MNAR)**. A woman with a low offered wage is more likely to drop out of the labour force or refuse to report a wage, i.e. her wage is missing precisely because of its value. This pattern of self-selection into reporting violates the MCAR and MAR assumptions and is exactly the type of problem the Heckman correction was designed for.

7. Three imputation methods for the wage regression

For each method I estimate

$$\text{logwage}_i = \beta_0 + \beta_1 \text{hgc}_i + \beta_2 \text{union}_i + \beta_3 \text{college}_i + \beta_4 \text{exper}_i + \beta_5 \text{exper}_i^2 + \eta_i.$$

The three columns of Table 2 correspond to (a) listwise deletion of complete cases, (b) mean imputation of `logwage`, and (c) the two-step Heckman selection model whose selection equation also includes `married` and `kids` (the exclusion restrictions).

Table 2: Returns to schooling under three missing-data treatments

	Listwise deletion	Mean imputation	Heckit (outcome)
(Intercept)	0.834***	1.149***	0.446***
	(0.113)	(0.078)	(0.122)
`hgc`	0.059***	0.036***	0.091***
	(0.009)	(0.006)	(0.010)
`unionYes`	0.222*	0.068	0.186*
	(0.087)	(0.047)	(0.084)
`collegeYes`	-0.065	-0.126**	0.092
	(0.106)	(0.048)	(0.100)
`exper`	0.050***	0.021**	0.054***
	(0.013)	(0.007)	(0.012)
`exper`^{2}	-0.004**	-0.001**	-0.002+
	(0.001)	(0.000)	(0.001)
Inverse Mills			-0.695***
			(0.060)
N	1545	2229	2229 (684 cens.)
R^2	0.038	0.020	0.092

$+\, p < 0.1$, $*\, p < 0.05$, $**\, p < 0.01$, $***\, p < 0.001$. Standard errors in parentheses. The Heckit selection equation also contains `married` and `kids`. See `regression_table.tex` for the raw `modelsummary` output.

Comparison of $\hat{\beta}_1$. The true return-to-schooling parameter is $\beta_1 = 0.091$. Across the three estimators we obtain

$$\hat{\beta}_1^{\text{listwise}} = 0.0590, \qquad \hat{\beta}_1^{\text{mean}} = 0.0363, \qquad \hat{\beta}_1^{\text{Heckit}} = 0.0915.$$

Both naive imputation methods produce a substantial *downward* bias. Listwise deletion underestimates the return by about a third, and mean imputation does even worse: replacing missing wages with the sample mean mechanically shrinks the variance of the dependent variable around its mean and pulls every slope toward zero, so $\hat{\beta}_1$ collapses to roughly 0.036. In contrast the Heckman two-step procedure recovers $\hat{\beta}_1 = 0.0915$, essentially identical to the true value of 0.091. The strongly significant inverse-Mills coefficient (-0.695, $p < 0.001$) confirms that the missing-wage process is non-random and that ignoring it produces biased OLS estimates.

Take-away. When data are MNAR, listwise deletion and mean imputation are both unreliable; mean imputation is especially dangerous because it injects fictitious observations at the centre of the distribution. Modelling the selection mechanism explicitly (Heckit) is the only one of the three methods that delivers an unbiased estimate of the returns to schooling here.

8. Probit model of union-job preferences

I estimate

$$u_i = \alpha_0 + \alpha_1 \text{hgc}_i + \alpha_2 \text{college}_i + \alpha_3 \text{exper}_i + \alpha_4 \text{married}_i + \alpha_5 \text{kids}_i + \varepsilon_i,$$

with `glm(family = binomial(link = "probit"))`. Estimates appear in Table 3.

Table 3: Probit model of union-job choice

	Estimate (s.e.)
(Intercept)	-6.743***
	(0.804)
`hgc`	-1.009***
	(0.098)
`collegeYes`	0.397
	(0.427)
`exper`	1.849***
	(0.156)
`marriedYes`	0.588**
	(0.206)
`kids`	0.799***
	(0.202)
N	2229

$**\ p < 0.01$, $***\ p < 0.001$. GLM warned of fitted probabilities very close to 0 or 1, suggesting the linear index is able to separate the two groups almost perfectly; the magnitudes of the coefficients should therefore be interpreted with caution.

9. Counterfactual: no penalty for being married or a mother

Following the procedure in `example.R`, I (i) compute predicted probabilities of union membership at the estimated parameters, (ii) zero out the coefficients on `marriedYes` and `kids`, and (iii) recompute the predicted probabilities. Averaging over the sample:

Scenario	Mean predicted Pr(union)
Baseline (estimated coefficients)	0.2373
Counterfactual ($\alpha_4 = \alpha_5 = 0$)	0.2174
Difference (counterfactual − baseline)	−0.0199

Because the estimated coefficients on `married` and `kids` are *positive*, removing them lowers the average predicted probability of choosing a union job by about 2 percentage points (from 23.7% to 21.7%). In other words, the model says that under the status quo, married women and mothers are *more* likely to take union jobs than their otherwise identical single, childless counterparts.

Is the model realistic? Probably not. Several features of the estimates are hard to defend.

1. *Sign of* `hgc` *and* `exper`. More schooling sharply lowers the probability of taking a union job ($\hat{\alpha}_1 = -1.01$), while more experience sharply raises it ($\hat{\alpha}_3 = 1.85$). Both magnitudes are implausibly large for a probit and the `glm.fit` warning about fitted probabilities of 0/1 is a classic sign of (near) complete separation – so the precise numbers cannot be trusted.

2. *No price variable.* We model the choice between a union and non-union job purely from demographics and human-capital variables. An economic model of job choice should at minimum include the union and non-union wage offers; the omission means the estimated $\hat{\alpha}$'s confound *preferences* with *offer probabilities*.

3. *Selection bias.* The sample only contains women who are employed, so we are estimating preferences conditional on labour-force participation, an issue we just demonstrated is non-trivial in Section 7.

4. *Counterfactual interpretation.* Even taken at face value, the policy "remove the union-job preference for marriage / motherhood" only changes the average probability by ≈ 2 percentage points. The estimated effect is therefore small and, given the issues above, not a credible basis for actual policy advice.

A more realistic specification would (a) include union and non-union wage offers (predicted from the Heckman wage regression), (b) jointly model labour-force participation and union choice, and (c) be estimated on a larger and more representative sample.

10–12. Repository synchronisation

The files `PS12_Cheng.R`, `PS12_Cheng.tex`, and `PS12_Cheng.pdf` were placed in `~/DScourseS26/ProblemSets/PS12` on OSCER and synced with the GitHub fork using the standard `git add / commit / push` workflow.